NC STATE Engineering

SESSION 5

Hands-On Activity: Building a Baseline ML Model

Dataset inspection • Baselines • Simple models • Metrics • Interpretation

Applied AI Design Workshop for Faculty

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Goal: complete one defensible first-pass ML workflow, not just run a model.

Activity outputs
- Problem statement
- Baseline model
- Trained simple model
- Initial metrics
- Interpretation and limitation statement

Session 5

From engineering problem to ML problem	Earlier Problem framing ML task types
Classic ML overview	
Invited presentation: From Prompts to Agents	
Break	Earlier Regression Classification
Regression fundamentals	
Classification fundamentals	
Hands-on baseline model activity	**Now**

Session 5 learning outcomes

Inspect
Check rows, variables, types, missing values, and target definition

Frame
Decide whether the dataset supports regression or classification

Baseline
Establish a simple comparison model before training anything complex

Train
Fit one or two simple models using a transparent workflow

Interpret
Compare metrics, identify limitations, and recommend a next step

Participant output: a concise model card for a first-pass baseline ML analysis.

Activity roadmap

Define → **Inspect** → **Baseline** → **Train** → **Evaluate** → **Interpret**

- **Define**: Target, inputs, task, metric
- **Inspect**: Rows, missing values, types, leakage
- **Baseline**: Mean/median or majority class
- **Train**: Simple model pair
- **Evaluate**: Metrics and error patterns
- **Interpret**: Limitations and next step

The purpose is to practice a complete, defensible workflow, not to optimize model performance.

Process Run Quality Dataset

A compact engineering dataset to practice the full workflow without domain-specific software barriers.

Run	Temp	Speed	Vibration	Material	Yield	Pass
1	185	42	0.18	A	88	Yes
2	190	46	0.35	B	72	No
3	174	39	0.22	A	91	Yes
4	205	52	0.48	C	61	No
5	198	44	0.20	B	86	Yes

Regression option
Predict continuous yield.

Classification option
Predict pass/fail quality.

Baseline required
Compare against simplest useful rule.

Part 1: define the problem

Start by writing down what the model is expected to predict and how success will be judged.

Decision or prediction

What question should the model answer?

Example: Will this run pass quality inspection?

Unit of observation

What is one row?

Example: one process run, part, student, machine cycle, sample, or day.

Metric

What score reflects success?

Example: recall if missed defects are costly; MAE if error magnitude matters.

one sentence problem statement + selected metric + baseline choice.

Part 2: inspect the data before modeling

A baseline model is only meaningful if the dataset has been checked for basic validity.

Rows and columns
Do we have enough observations for the number of variables?

Data types
Which features are numerical, categorical, date/time, or text?

Missing values
Are values missing randomly, systematically, or by design?

Target definition
Is the target measurable, consistent, and available after the prediction point?

Leakage risk
Would any input reveal the answer too directly or too late?

Split strategy
Should the data be split randomly, by group, or by time?

Mini-lesson: data leakage can invalidate results

Leakage occurs when the model uses information that would not be available at the time of prediction.

Example leakage feature

Final inspection score included as an input when predicting pass/fail quality.

The model is effectively seeing the answer.

Time leakage

Using future sensor readings when predicting an earlier failure or yield.

The timeline is broken.

Group leakage

Samples from the same subject, machine, or batch appear in both train and test sets.

The test set is too familiar.

Activity: Identify one variable that could cause leakage and justify keeping or removing it.

Part 3: establish the baseline

The baseline answers: what performance can we get with a simple, transparent rule?

Regression baseline

Predict the training-set mean or median.

Metrics: MAE, RMSE, R^2 relative to simple prediction.

Classification baseline

Predict the majority class or current rule.

Metrics: confusion matrix, accuracy, precision, recall, F1.

Why it matters

A model that does not beat the baseline should not be treated as useful, even if it is complex.

Part 4: train simple models

Use simple, interpretable models first. The goal is to understand the problem before chasing accuracy.

Regression path

1. Mean baseline
2. Linear regression
3. Decision-tree regression

Compare training and test error.

Classification path

1. Majority baseline
2. Logistic regression
3. Decision tree

Compare confusion matrices.

Model discipline

Use the same split, same inputs, and same target for each comparison.

Change one thing at a time.

Part 5: evaluate results

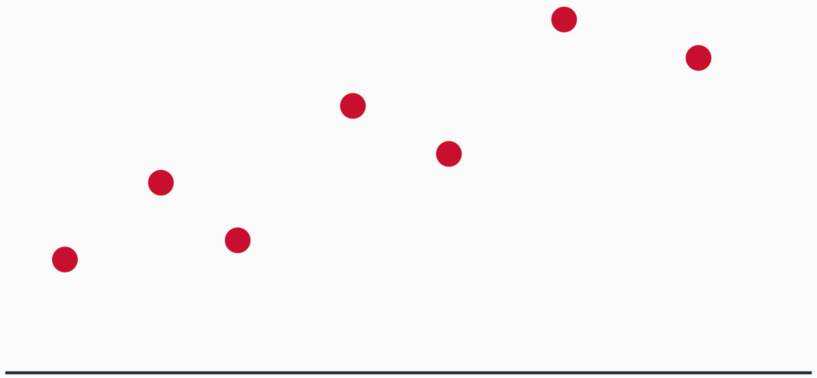
Evaluation should answer both “How well did it predict?” and “What kinds of errors did it make?”

Regression	MAE	Average absolute error in original units
Regression	RMSE	Penalizes larger errors more strongly
Regression	R^2	Fraction of target variation explained
Classification	Accuracy	Overall percent correct
Classification	Recall	Fraction of important positives detected
Classification	Precision	Fraction of predicted positives that were correct

Metric choice should reflect the practical consequence of the error, not just mathematical convenience.

Part 5 continued: inspect error patterns

A first model is most valuable when it shows where the data or problem formulation needs improvement.



Residuals

TP	FN
FP	TN

Questions

- Are errors larger for one group?
- Are mistakes tied to missing features?
- Is one class poorly represented?
- Is the target noisy?

Error analysis converts model performance into research and data-collection priorities.

Participant output: first-pass model card

Each group should leave with a short model card summarizing the baseline analysis.

Problem statement
What is predicted, for whom or what, and why?

Dataset
Rows, variables, target, missing values, and leakage concerns

Baseline
Simple rule and its performance

Model
Algorithm, inputs, split method, and metric

Results
Performance compared with baseline and observed error pattern

Limitations
What should not be concluded and what data are needed next?

Discussion prompts

- Did the model outperform the baseline?
- Was the improvement practically meaningful?
- Which metric best represents success?
- Did the model appear to overfit?
- Were any variables likely to introduce leakage?
- Would the model generalize to another lab, machine, course, or population?
- What additional data should be collected?
- What limitation must be reported?

Synthesis: a baseline workflow is a scientific argument about whether the data support a useful prediction.

Session 5 wrap-up: what you should take forward

A complete first-pass ML workflow should be transparent, reproducible in class, and easy to critique.

Before modeling

- Define target
- Define unit
- Select metric
- Inspect data
- Check leakage

During modeling

- Establish baseline
- Use simple models
- Keep comparison fair
- Evaluate on held-out data

After modeling

- Interpret errors
- State limitations
- Recommend next data
- Avoid causal overclaims

Day 3 endpoint: You can frame an applied ML problem and run a defensible baseline analysis.